



Expressing an arithmetic sequence

Task A: The position of a term in a sequence

1) Match the term number (n) to the term values (T). One has been done for you.

a)

Term number (n)	1	2	3	4	5
Term value (T)	2	5	8	11	14

$n = 1$

$T = 14$

$n = 2$

$T = 2$

$n = 3$

$T = 8$

$n = 5$

$T = 5$

b)

Term number (n)	1	2	3	4	5
Term value (T)	5	4	3	2	1

$n = 1$

$T = 1$

$n = 2$

$T = 4$

$n = 3$

$T = 2$

$n = 4$

$T = 3$

$n = 5$

$T = 5$

2) Andeep and Jacob are looking at an arithmetic sequence.

Term number (n)	1	2	3	4	5
Term value (T)	45	40	35	30	25

I see the pattern!
When $n = 0$ the term is 50

I agree! Next would be
when $n = -1$
The term will be 55

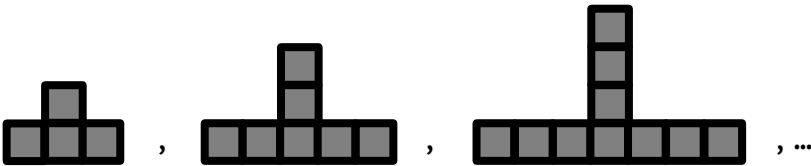


Write a sentence explaining why Andeep and Jacob are wrong.



Task B: The general term of a sequence

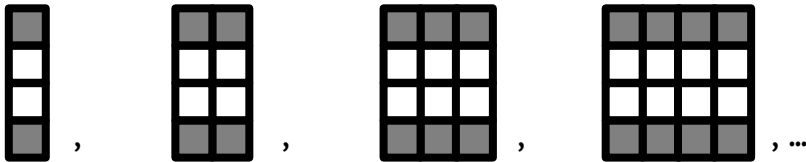
1) Sam spots a general rule for this sequence.



Pattern number	1	2	3	Pattern number
Squares	4	7	10	Pattern number $\times 3 + 1$

Whilst their general rule works, how would you improve upon it?

2) Fill in the table for this sequence and write a general rule each row.



Pattern number (n)	1	2	3	4	5				n
Shaded squares	2					50			
Total squares		8					200		
Perimeter	10							208	